Polarization: Verification of the Malus' Law and a Slight Touch on the Quarter Wave Plate Model

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This experiment aimed to test the validity of Malus' law and investigate the behavior of light passing through a quarter-wave plate in a Junior Lab environment. It involved the use of a polarizer to produce linearly polarized light, which was then passed through various optical instruments. A lock-in-amplifier and a beam chopper are used together to measure the intensity of transmitted light from analyzer. The results confirmed the cosine squared relationship predicted by Malus' law, providing strong empirical support for the law's validity. Furthermore, the experiment examined the effect of a quarter-wave plate on polarized light. By fixing the fast or slow axis of the $\lambda/4$ plate in line with the incident polarization direction, we reproduced the results as expected by theoretical predictions of the Malus' law.

1. INTRODUCTION

The phenomenon of polarization affirms the transverse nature of electromagnetic waves [6]. Electromagnetic waves are polarized if their electric field vectors are all in a single plane, called the plane of oscillation. Light waves from common sources like light bulb are not polarized; that is, they are unpolarized, or polarized randomly. Conversely, light from a rainbow, reflected sunlight, and coherent laser light are examples of polarized light. There are three different types of polarization states: linear, circular and elliptical. Each of these commonly encountered states is characterized by a differing motion of the electric field vector with respect to the direction of propagation of the light wave. When a polarizing sheet is placed in the path of light, only electric field components of the light parallel to the sheet's polarizing direction are transmitted by the sheet; components perpendicular to the polarizing direction are absorbed. The light that emerges from a polarizing sheet is polarized parallel to the polarizing direction of the sheet[5]. Polarization implies sidedness. Malus discovered that the elemental constituents of light were somehow sided. This bias was revealed only after more than a century during which the scientists strained at the doubled images produced by calcite.

It's only in 1808, when Étienne-Louis Malus (1775–1812) gazed through a doubly refracting crystal(calcite) at light reflected from the windows of the Luxembourg Palace in Paris, expecting to see two equally bright images. Rather, one of the transmitted rays was brighter, and their relative intensities depended on the position of the crystal. Malus initially tried to explain this anomaly by supposing that the atmosphere somehow modified the solar light [1]. Later that evening he let a taper of candlelight reflect off the surface of a basin of water. Analyzing this reflected light with the same calcite crystal, he observed the characteristic intensity changes. Malus realized then that not only light emerging from a doubly refracting crystal can subsequently lose its ability to constantly divide itself into two rays, but also that reflected light can behave similarly [7].

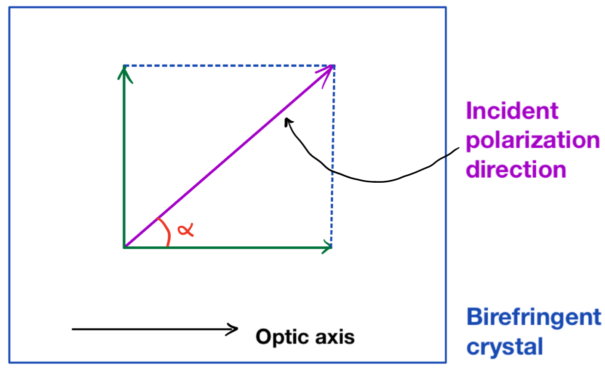

FIG. 1. Splitting of linearly polarized light into two components as it enters a birefringent crystal.

This observation became firmer following a series of investigations of the angular dependence of the reflection of light already polarized by another glass surface. The device Malus constructed for these measurements was the first polarimeter. With this apparatus, Malus deduced his eponymous law. "The quantity of light", he wrote, "reflected by [the second glass at polarizing incidence] is proportional to the square of the cosine of the [torsion] angle ranging between the [pivots of the two glasses]..." [9]. Malus' Law apart from being the first quantitative relationship for treating polarized light intensities, showed that plane-polarized light can be represented by a vector that can be resolved into components as can any other vector[4].

A retarder (waveplate) is a birefringent crystal that resolves a light wave into two orthogonal linear polarization components by producing a phase shift between them as shown in Fig.1. Depending on the induced phase difference, the transmitted light may have a different type of polarization than the incident beam. The wave plates do not polarize unpolarized light and ideally, they do not reduce the intensity of the incident light beam.

A quarter-wave plate is employed in this experiment which is used to convert linear polarization to circular polarization and vice-versa. In order to obtain circular polarization, the amplitudes of the two electric field vector components must be equal and their phase difference must be 90° or 270°. A quarter-wave plate is capable of introducing a phase difference of 90° between the two components of incident light. Moreover,the direction of polarization of the incident light with respect to the optic axis of the quarter wave plate is equally important to produce circular polarization. From Fig.1, it is obvious that the amplitudes of the two components will be equal only when the incident linear polarization direction makes an angle of 45° with respect to the optic axis of the crystal (i.e. $\alpha = 45°$). Note that contrarily, if circularly polarized light is incident on a quarter-wave plate, the resulting polarization will be linear, at an angle of 45° with respect to the optic axis. Again, the reason is that a phase difference of 90° is introduced between the two components of the electric field vector, which traces a helix-like path as the circularly polarized wave propagates.

Malus' discoveries of the production and behaviour of polarized light were responsible for the development of the wave theory of light[3], the science of molecular chirality[8]. In the 21st century, some of the strangest phenomena in nature and the most fantastic technologies — nonlocality and quantum computing, respectively rely on entangled, polarized photons[2]. This experiment aimed to verify the validity of Malus' law and the model of a quarter wave plate.

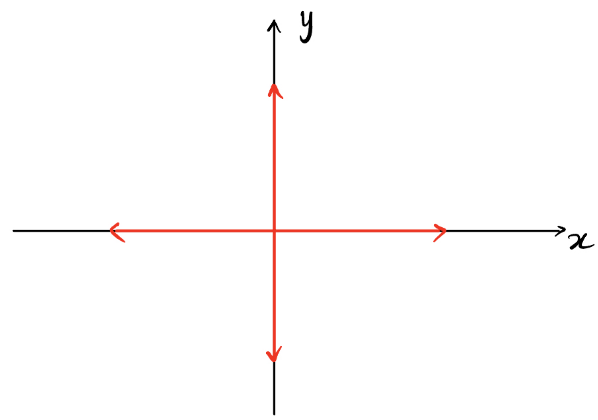

FIG. 2. A quick way of representing unpolarized light — the light is the superposition of two polarized waves whose planes of oscillation are perpendicular to each other.

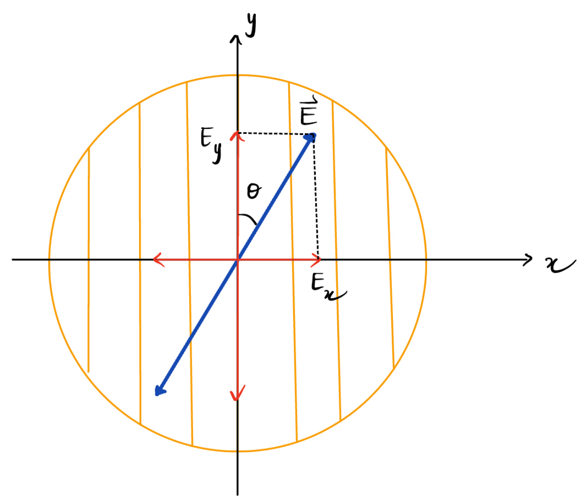

FIG. 3. Polarized light approaching a polarizing sheet. The electric field $\vec{E}$ of the light can be resolved into components E_y (parallel to the polarizing direction of the sheet) and E_x (perpendicular to that direction). Component E_y will be transmitted by the sheet; component E_x will be absorbed.

2. THEORY

Consider an unpolarized light, whose electric field oscillations can be resolved into x and y components approaches a polarizing sheet as represented in Fig.2. Further, we can arrange for the y axis to be parallel to the polarizing direction of the sheet. Then only the y components of the light's electric field are passed by the sheet; the x components are absorbed. As suggested by Fig.2, if the original waves are randomly oriented, the sum of the x components and the sum of the y components are equal. When the x components are absorbed, half the intensity I_0 of the original light is lost. The intensity I of the emerging polarized light is then

$$I = \frac{I_0}{2} \tag{1}$$

2.1. Linearly polarized light

Suppose now that the light reaching a polarizing sheet is already polarized. Fig.4 shows a polarizing sheet in the plane of the page and the electric field $\vec{E}$ of a polarized light wave travelling toward the sheet. Resolving $\vec{E}$ into two components relative to the polarizing direction of the sheet: parallel component E_y is transmitted by the sheet, while the perpendicular component E_x is absorbed. Since θ is the angle between $\vec{E}$ and the polarizing direction of the sheet, the transmitted parallel component is

$$E_y = E cos\theta \tag{2}$$

But the intensity of an electromagnetic wave (such as our light wave) is proportional to the square of the electric field's magnitude ($I = E_{rms}^2/c\mu_0$). Here, the intensity I of the emerging wave is proportional to E_y^2 and the intensity I_0 of the original wave is proportional to E^2. Hence, from Eq.2 we can write

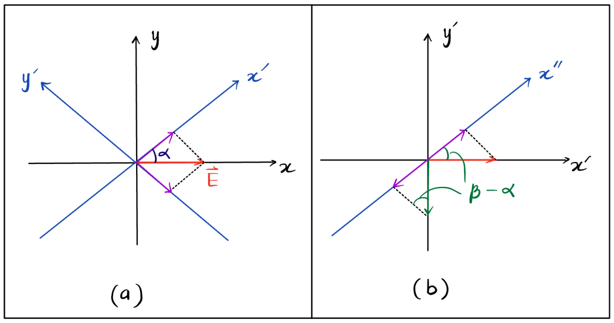

FIG. 4. (a) The E vector incident on the quarter wave plate decomposes into components parallel and perpendicular to the optic axis. (b) The components along the transmission axis of the second linear polarizer are transmitted.

$$I = I_0 cos^2\theta \tag{3}$$

Eq.3 is the mathematical representation of Malus' law. Note that the transmitted intensity I is a maximum and is equal to the original intensity I_0 when the original wave is polarized parallel to the polarizing direction of the sheet (when θ in Eq.3 is 0° or 180°). Also, the transmitted intensity is zero when the original wave is polarized perpendicular to the polarizing direction of the sheet (when θ is 90°).

The above analysis assumes a totally linearly polarized light source. Note that if unpolarized or partially polarized light is used, the transmission function takes the form,

$$I = Acos^2\alpha + B \tag{4}$$

where A and B are constants.

2.2. Circularly polarized light

Consider the totally linearly polarized light emerging from the first linear polarizer. Now, lets place a $\lambda/4$ plate between the first polarizer and the second polarizer (the analyzer). Fig.4(a) shows the x-axis along the transmission axis of the first polarizer and the x'-axis along the optical axis of the quarter-wave plate. The angle between the x-axis and the x'-axis is α. Decomposing the electric field vector $\vec{E}$ along the x-axis into $E_x = Ecos\alpha$ along x' and $E_y = Esin\alpha$ along y'. The $\lambda/4$ plate shifts the phase of the component along x' by $\pi/2$ radians. Assuming that the incoming light is represented in complex notation as $Ee^{i\omega t}$, the components along x' and y' are,

$$\begin{aligned} E_x &= Ee^{i\omega t + \frac{\pi}{2}} cos\alpha \\ E_y &= Ee^{i\omega t} sin\alpha \end{aligned} \tag{5}$$

(Note that $e^{i\frac{\pi}{2}}$ represents a $\pi/2$ phase shift)

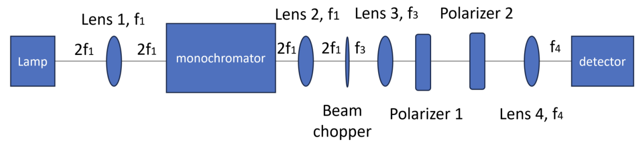

FIG. 5. Schematics of general optical setup, showing the distances $2f_1$, f_3, and f_4 between different optical instruments.

Notice that the magnitude of $\vec{E}$ that results after the quarter-wave plate is still E, and therefore the transmission function of the quarter wave plate is just $I = I_0$ where I is the intensity after the quarter-wave plate and I_0 is the incident intensity. In Fig.4(b), the components along the x' and y' axes are projected onto the x"-axis, which represents the transmission axis of the analyzer. The x" makes an angle β with the x-axis, or an angle $(\beta - \alpha)$ with the x'-axis. The resulting E' field is,

$$E' = Ee^{i\omega t}[e^{i\frac{\pi}{2}} cos(\alpha).cos(\beta-\alpha) - sin(\alpha).sin(\beta-\alpha)] \tag{6}$$

But the intensity I of the transmitted light is proportional to $|E'|^2$, i.e., $I \propto |E|^2 = E'.\bar{E'}$. Where $\bar{E'}$ is the complex conjugate of E'. This yields

$$I \propto E^2[cos^2(\alpha).cos^2(\beta-\alpha) + sin^2(\alpha).sin^2(\beta-\alpha)] \tag{7}$$

Eq.7 is the transmission function of the second linear polarizer if circularly polarized light is incident on it. Substituting $\alpha = 45°$ into Eq.7 gives back Eq.1, i.e., $I = I_0/2$.This verifies that circular polarization should be obtained by placing the optic axis of a quarter wave plate at 45° with respect to an incident linearly polarized light. We also conclude that rotating the second linear polarizer does not change the transmission of the circularly polarized light.

3. EXPERIMENT

This experiment employed various physical system apparatus such as a mercury discharge lamb, linear polarizers, lenses, monochromators, and quarter wave plates. In addition, the measurement equipment consists of a lock-in amplifier, beam chopper, and a photodiode to measure the signal (light intensity). A schematics experimental setup for this experiment is shown in Fig. 5.

Given the wide range of wavelength available at the input, a convex lens (with focal length $f_1 = 15$ cm) was used to focus the light on the monochromator, which then transmit a mechanically selectable narrow band of wavelengths of lights. Choosing green light due to its clarity even with room lights on, we focused it onto the chopper blades using the second lens (with focal length $f_2 = 15$ cm) to properly modulate the signal with the beam chopper. The beam chopper, lock-in-amplifier, and photodiodes are used together to measure the signal (light intensity). In addition to modulating the signal at a set rate,

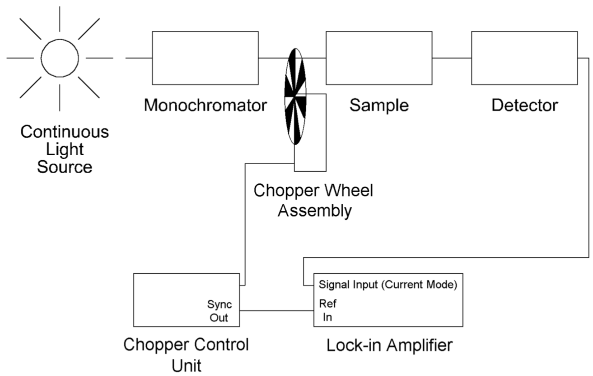

FIG. 6. Schematics of the signal measurement system using a Lock-in amplifier and beam chopper. The lock-in-amplifier takes a reference signal from the beam chopper and a signal from the light detector to produce the output signal.

the beam chopper provides a synchronous reference signal capable of driving the reference channel of a lock-in amplifier. The beam chopper consisted of a control unit and a chopper wheel. We selected the chopping frequency as 170 Hz, which is much higher than the frequency of the electrical signal (60 Hz), but not too high to cause aliasing or distortion. Since the rotation of the blade causes the light signal path to be interrupted, the light source that stimulates the experiment is in the form of AC excitation. Since the aperture size is large compared to the beam diameter, this excitation is an optical equivalent of a square wave. So, we chose the reference output voltage as square wave on the lock-in amplifier.

Any discrete frequencies or noise voltages not equal to the reference frequency was rejected by the lock-in amplifier. The result was a much lower limit on signals which can be measured. This technique allows measurements to be made with the room lights on as long as the magnitude is insufficient to saturate the detector. Still then, we insured that stray light does not enter the experiment via the chopped light path. Fig. 6. depicts schematics of the signal measurement system using a Lock-in amplifier and beam chopper.

Trying our very best to keep the path of the light straight, we then focussed the light from beam chopper onto the linear polarizer using the third lens (with focal length $f_3 = 10\ cm$). We used the linear polarizers which transmit a beam of light whose electric field vector oscillates in a plane that contains the beam axis. Upon passing through the analyzer, the light was then focused onto the photo detector using the fourth lens (with focal length $f_4 = 7.5\ cm$). It is to be noted that since we used photodiode as light detector, the light intensity determines the current generated by the photodiode, which is then converted to voltage in Lock-in amplifier. Because of the low power (intensity of the light is below the saturation point of the photodiode), the intensity of the light detected by the photodiode is directly proportional to the voltage reading on the Lock-in amplifier. For this reason, we took the voltage reading on the Lock-in amplifier as intensity of the light. As explained above the signal from the light detector was fed into lock-in amplifier which together with beam chopper, produces an output signal.

3.1. Malus' Law

To work with an offset from 0°, we kept the polarizer fixed and rotated the analyzer until we observed a maximum intensity on the lock-in amplifier. A maximum intensity of around 198.6 mV was observed at $\theta_0 = 102°(= 0°)$. At that point the pass axes of the polarizer and the analyzer are parallel. We then rotated the analyzer in 15° increment from 102° to 460° (i.e., 0° to 360°) and took the reading of the intensity at each angle. To do a further analysis of the Malus' law, we took a reading of intensity from 0° to 90° with an increment of 5° (we could have chosen 90° - 180°, 180°- 270° or 270°-360° range). Although we can extract this data from prior data collected, we will only get six (90 / 15) measurements which is probably a not good sample size. With this data set, we intended to plot a graph of intensity I vs. $cos^2\theta$. We expect to get a straight line with its slope equal to the initial intensity I_0 if Malus' law is valid.

3.2. Quarter Wave Plate Model

By setting the relative difference of angle between two polarizers as 0°, we inserted the quarter waveplate between two polarizers and gradually rotated it around its axis while monitoring the intensity of the transmitted light. Recording a maximum intensity on the Lock-in amplifier suggests that the axis of the polarized light is in line with either fast or slow axis of the quarter wave plate. Luckily, we observed that the maximum intensity occurred at 0 ± 1.5°. An uncertainty of 1.5 degree was taken because we used a bold mark on the protractor scale of the quarter wave plate. A schematics of the optical setup of the waveplate that is modelled is shown in Fig. 7.

Keeping relative difference of angle between two polarizers fixed as 0°, we then rotated the quarter wave plate to few prominent angles such as 0°, 90°, 180°, and 270° and recorded the intensity. Since these angles corresponds to either fast or slow axis of the quarter wave plate, we should be expecting the no change in intensity and still get the maximum intensity measurement. By changing the relative difference of angle between two polarizers to 90°, we repeated the previous step expecting to get a minimum intensity at each step. Clearly, if we measure the intensity of the transmitted light as function of the angle of analyzer in this setting, it is expected to follow Malus' law.

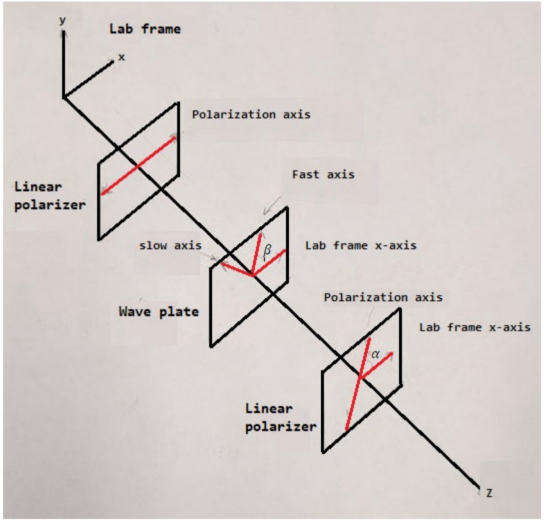

FIG. 7. Schematics of optical setup of the waveplate showing the angle between the incident polarization direction and the fast axis of the wave plate β, and the angle between the initial polarization state of the beam and the axis of the analyzer. Both angles are depicted with respect to the lab frame.

Satisfied with our result, we then fixed the angle between the incident polarization direction and the fast axis of the wave plate β to 45°. At this angle, the quarter waveplate is oriented such that it transforms linearly polarized light into circularly polarized light. In this setup, we expect a constant intensity of the transmitted light when we vary the angle of the analyzer keeping the relative angle of waveplate with respect to polarizer fixed as 45°. To our dismay, we could not obtain these relationship. One possible reason might be because the range of intensity measurement on the Lock-in amplifier got widespread and it hardly settled down to a reasonable range. Insofar, I have only covered some basics on waveplate model in this report, but more data is required to draw a rigorous and constructive conclusions.

4. RESULTS AND DISCUSSION

4.1. Malus' Law

We observed that the intensity measurement hardly settles down to a fixed value. So, we took the average of highest and lowest intensity as our measurement. Moreover, the difference between highest and lowest intensity remains constant for all the angles. Thus, I took the average of difference between highest and lowest intensity as the estimated uncertainty in the intensity measurement. I performed two least square fitting using Python to verify the validity of Malus' law. Fig.8 shows a cosine-squared curve fitted to the experimental data of the intensity measured at different angles from 0° - 360°, with a step size of 15°. Likewise Fig.9 shows the same curve fitted to the experimental data of the intensity measured at different values of $cos^2\theta$ from 0 to 1, with a step size of 0.05.

Assuming that each intensity measurement I_j is normally distributed about the model equation $I_0 cos^2\theta_j$, with width parameter σ_I leads to say the deviations $I_j - I_0 cos^2\theta_j$ are normally distributed, all with central value zero and the same width σ_I. This situation immediately suggests that a good estimate for σ_I would be given by root mean square of $I_j - I_0 cos^2\theta_j$, which is the deviation of model function from the measured values. Applying this principle, the standard deviation $\sigma_I = 9.6$ mV is obtained using Python for Fig.8. Hence, the uncertainty of each measured intensity δI_j is 2.0 mV. The least squared estimate of the initial intensity $I_0 = 190.4 \pm 2.0$ mV.

Fig.9 is the result of the analysis of the fitting data to the linear equation $y = ax + b$, by setting the light intensity I as the dependent variable y and $cos^2\theta$ as the independent variable x. The coefficient of determination $R^2 = 0.999$ for the regression line of I on $cos^2\theta$ indicates a strong positive correlation between I and $cos^2\theta$ as expected from the Malus' law equation. Though the intensity at 90° (or 270° in prior case) were expected to be zero ideally, the lowest intensity we observed was 0.2 mV. This might be contributed by the stray light (room light) detected by the photo detector. Just as above, the standard deviation $\sigma_I = 0.97$ mV was obtained for Fig.9. Hence, the uncertainty of each measured intensity δI_j is 0.23 mV. The least squared estimate of the initial intensity $I_0 = 186.26 \pm 0.23$ mV.

TABLE I. Intensity measured for varying angle between the incident polarization direction and the fast/slow axis of the wave plate β from 0° to 360° with the angle of analyzer measured with respect to angle of polarizer θ fixed as 0° and 90°.

β / ± 1.5°	$\theta = \alpha - \alpha'$ / degrees	Intensity I / ± 2.2 mV
0	0	182.9
90	0	177.4
180	0	172.1
360	0	179.2
0	90	0.2
90	90	0.3
180	90	0.4
360	90	0.2

4.2. A Slight Touch on Quarter Wave Plate Model

Table.I depicts the intensity measured for varying angle of waveplate β keeping the angle of analyzer θ fixed as 0° and 90°. For $\theta = 0°$, the measured intensities

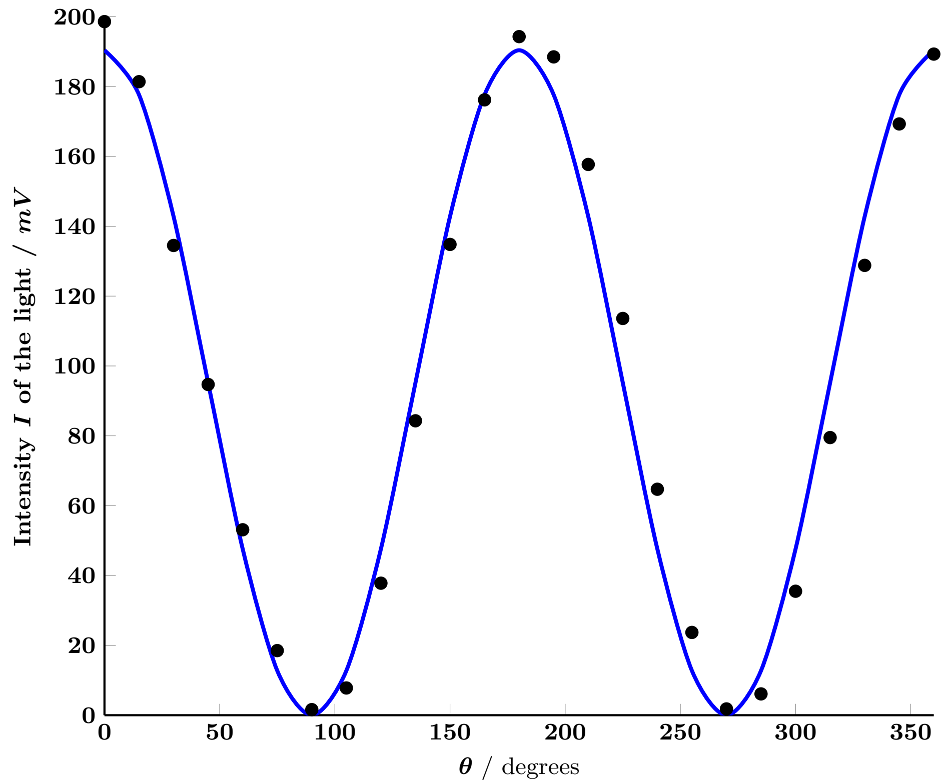

FIG. 8. Plot of the intensity I of the light after transmitting through the analyzer as a function of angle θ of Eq.3. The initial velocity I_0 value determined from the least square fitting of the data was $I_0 = 190.4 \pm 2.0\ mV$.

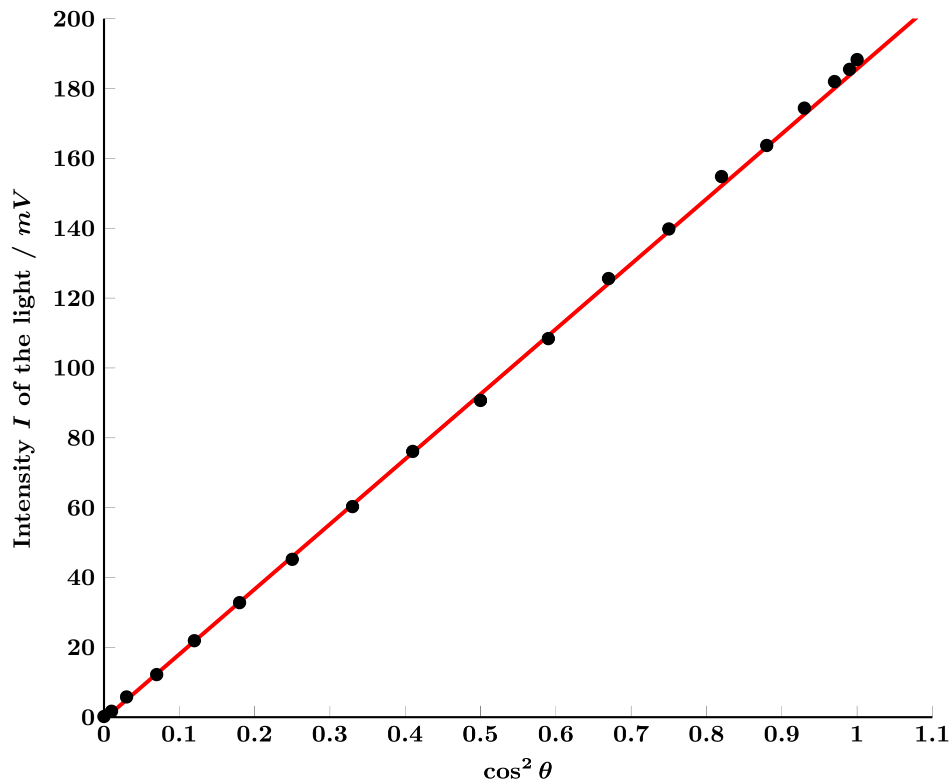

FIG. 9. Plot of the intensity I of the light after transmitting through the analyzer as a function of $cos^2\theta$ of Eq.3. The initial velocity I_0 value determined from the least square fitting of the data was $I_0 = 186.26 \pm 0.23\ mV$.

were very close to the maximum intensities in part.4.1. This confirms the β's considered corresponds to either fast or slow axis of the quarter wave plate. Similarly, for $\theta = 90°$, the measured intensities were very close to the minimum intensities in part.4.1. Taking an average of the measured intensities and standard deviation using excel LINEST function, we get $I_{max} = I_0 = 178.0 \pm 2.2\ mV$ and $I_{min} = 0.3 \pm 0.05\ mV$ for $\theta = 0°$ and $90°$ respectively. Evidently, the quarter wave plate has no effect on the intensity of light when $\theta = 0°$ or $\theta = 90°$.

A least square fitting done for two sets of data in part.4.1 resulted in an initial intensity I_0 of 190.4 ± 2.0 mV and $186.26 \pm 0.23\ mV$. We got $I_0 = 178.0 \pm 2.2\ mV$ from the part.4.2. Clearly, despite their close proximity, the three values are off from each others uncertainty. One obvious reason is improper functioning of the Lock-in amplifier used during the experiment. Besides fluctuating on a broad range of intensity values for each angle, the measurement values for a same angle kept dropping over time. Furthermore, errors in measuring the precise angles might have also contributed to inaccuracies in the intensity measurements. Also, polarizers might have non-zero extinction ratios.

Still then some apparatus revision could be made to further improve and reduce the associated uncertainty in the measurements. Having a better Lock-in amplifier might aid to reducing the uncertainty of I_0. Though the use of lock-in amplifier and beam chopper allows one to do this experiment with room lights on, performing the experiment in dark room (with stray lights) might help in getting a more precise intensity measurement. Performing the experiment in dark room means we can study the intensity of pale lights too (we focused on green light because it can be distinctly seen even with room lights on). Covering the photodetector to shield it from ambient light can be more effective rather that risking one's safety by working in dark room. Some model revisions can be made by adding a fit variable to Malus' law to represent offsets due to ambient light or determining the non-linear calibration of the photo detector for high light intensities.

5. CONCLUSIONS

The first part of this experiment attested the validity of Malus' law, thus proving that the intensity of light transmitted through analyzer is directly proportional to the cosine squared of the angle between the initial polarization state of the beam and the axis of the analyzer. Malus' law finds practical applications in diverse fields such as designing polarized sunglasses, optimizing liquid crystal displays, and analyzing polarization properties in scientific research.

The second part of this experiment demonstrates that the quarter wave plate acts like a glass and does not effect the intensity of incident polarized light when the incident polarization direction aligns with either fast or slow axis of the quarter wave plate. Thus, measuring the intensity at various angles of analyzer followed Malus' law as expected. Further analysis can be done about conversion of linearly polarized light to circularly polarized light and elliptically polarized light by quarter waveplate. This conversion of linearly polarized light into circularly polarized light and otherwise enable applications such as optical rotation measurements and optical isolation.

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